Analytical Chemistry
Cumulative Exam
December 3, 2010

1. (5 points) Consider a scenario where you are reading a variety of papers about materials chemistry. In these papers some researchers discuss using ESCA and others use XPS for different analyses. What is the difference between these two spectroscopic techniques?

2. (5 points) LIBS and other spectroscopic techniques often use a CCD camera as a detector. What is a CCD camera? How does it work? Use diagrams if necessary.

3. (5 points) What elements are not detectable by XPS?

4. (5 points) What is a TOF analyzer and by what physical or chemical principles does it separate ions?

5) (80 points) For <u>each</u> of the following spectroscopic techniques listed below, provide the following information that is categorized below.

<u>Techniques.</u>

a) AES

b) XPS

c) LIBS

d) TOF-SIMS

<u>Information to include in your discussion for each of the techniques.</u>

i) Write the full name for the technique. For example, ESCA = Electron Spectroscopy for Chemical Analysis.

ii) In your own words, <u>briefly</u> describe how the instrumental technique works. Be sure to include appropriate chemical or physical principles in your discussion.

iii) Sketch a diagram of the instrumental set up.

iv) Provide one example of a chemical analysis problem that could be solved using the spectroscopic technique. Please provide a different scenario for each technique (4 different chemical analysis applications). Be sure to indicate if any of the other techniques listed in the list could also be used to solve the chemical analysis proposed. For example, if you provide an example for LIBS and TOF-SIMS can be also applied to gain similar chemical information, you must state this in your answer.